An explicit isoperimetric constant for Thompson's group F

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Abstract

The recent preprint [1] proves that Thompson's group F is nonamenable, by a finite averaging argument driven by a Lipschitz self-map of the Hilbert unit ball with displacement bounded away from zero; the argument is formalized in Lean. Neither the preprint nor the formalization gives an explicit boundary constant or a constant for a prescribed generating set. We make the argument explicit for the standard generators x_0, x_1. With

$$h_E(F) = \inf_{\emptyset \neq A \subset F \text{ finite}} \frac{|x_0 A \triangle A| + |x_1 A \triangle A|}{|A|}$$

(the Cheeger constant of the 4-regular Cayley graph), we prove unconditionally that $h_E(F) \geq 2.109 \times 10^{-19}$, and $h_E(F) \geq 2.466 \times 10^{-19}$ with a sharper word count. If the closed unit ball of an infinite-dimensional Hilbert space admits a k-Lipschitz retraction onto its sphere, then $h_E(F) \geq g(k)$ for an explicit function g; the value $k_0(\ell^2) \leq 28.99$ reported in the literature (we have not seen the original source) would give $h_E(F) \geq 8.596 \times 10^{-9}$. The ingredients are a computer-assisted evaluation of one separation constant ($\sigma(1/32) = 1/256$), the Hilbert-space form of the normalization estimate, and direct word-length bounds for the transports in Cannon–Floyd–Parry normal form. This is a standard effectivization of [1]; we add no new mechanism. The best known upper bound is $h_E(F) < 0.4996$ (Guba).

1 Introduction

Let F be Thompson's group of increasing piecewise linear homeomorphisms of $[0,1]$ with dyadic breakpoints and slopes in $2^{\mathbb{Z}}$. We write products as composition, $hg = h \circ g$, as in [1], and use the standard generators

$$x_0 : \ [0, \tfrac{1}{2}] \to [0, \tfrac{1}{4}], \ [\tfrac{1}{2}, \tfrac{3}{4}] \to [\tfrac{1}{4}, \tfrac{1}{2}], \ [\tfrac{3}{4}, 1] \to [\tfrac{1}{2}, 1];$$
$$x_1 = \text{id on } [0, \tfrac{1}{2}], \text{ a half-scale copy of } x_0 \text{ on } [\tfrac{1}{2}, 1].$$

For a nonempty finite $A \subset F$ put $\eta_i(A) = |x_i A \triangle A|/|A|$ $(i = 0, 1)$. In the Cayley graph with edges $g \sim x_i^{\pm 1} g$, a 4-regular graph, the number of edges leaving A is $\sum_{s \in \{x_0^{\pm 1}, x_1^{\pm 1}\}} |sA \setminus A| = |x_0 A \triangle A| + |x_1 A \triangle A|$, because $|x^{-1} A \setminus A| = |A \setminus xA|$ and $|xA \setminus A| = |A \setminus xA|$. So

$$h_E(F) = \inf_A \big(\eta_0(A) + \eta_1(A)\big)$$

is the edge-isoperimetric (Cheeger) constant of this graph. The map $g \mapsto g^{-1}$ identifies the graph with the right Cayley graph, so $h_E(F)$ equals the constant ι^* of Guba [9], which satisfies $\delta + \iota^* = 4$ with δ the density. By Følner's criterion, F is nonamenable if and only if $h_E(F) > 0$.

*The results, proofs and computer certificates in this note were found and checked by Claude (Anthropic) agents working under the direction of [AUTHOR NAME]. The note has had no human peer review.

Theorem 1.1 (unconditional)**.** *For every nonempty finite $A \subset F$,*

$$|x_0 A \triangle A| + |x_1 A \triangle A| \;\geq\; c_\star\,|A|, \qquad c_\star = 2.1091\ldots \times 10^{-19}.$$

More precisely, $c_\star = (\kappa - 4/D)/(4(1-1/D)\bar{M})$ with $\kappa = (2 \cdot 29621120)^{-2}$, $D = 2^{59}$ and $\bar{M} = \bar{M}(59) = 329.5 + 2 \cdot 59/(2^{59}-1)$ given in closed form in Lemma 5.4. The sharper word count of Lemma 5.3 gives $h_E(F) \geq 2.4666 \times 10^{-19}$ (Theorem 6.1).

Theorem 1.2 (conditional on a retraction constant)**.** *Suppose that for some $k > 1$ the closed unit ball B of an infinite-dimensional real Hilbert space admits a k-Lipschitz retraction onto its unit sphere. Then $h_E(F) \geq g(k)$, where*

$$g(k) = \max_{m \geq 1} \frac{\kappa_k - 4 \cdot 2^{-m}}{4(1-2^{-m})\bar{M}(m)}, \qquad \kappa_k = \Big(\frac{2}{k(k+2)}\Big)^2,$$

and $\bar{M}(m)$ is the explicit average word length of Lemma 5.4. Values are in Table 1 and Figure 1; in particular $g(29) = 8.5964 \times 10^{-9}$, $g(32.26) = 5.540 \times 10^{-9}$, and $g(28.99) = 8.6085 \times 10^{-9}$.

The optimal retraction constant $k_0(H)$ of a Hilbert space is known to satisfy $4.5 \leq k_0(H) \leq 28.99\ldots$. Chaoha–Goebel–Termwuttipong [7] quote $4.5\ldots \leq k_0(H) \leq 28.99\ldots$ as the published estimates "(see books and [2])"; Mahesh [12] quotes $4.58 \leq k_0(\ell^2) \leq 28.99$ citing Casini–Piasecki [10] and [2]. The upper bound may therefore come from [2] or from [10]; we have read neither. *We have not been able to read the full text of [2].* So we have not checked that the bound there is a proved theorem rather than a numerical estimate, or how many digits it has. Since all separable infinite-dimensional real Hilbert spaces are isometric to ℓ^2, if the bound holds as quoted then some retraction has constant < 29, and Theorem 1.2 gives $h_E(F) \geq g(29) = 8.596 \times 10^{-9}$. We state this consequence only conditionally. The value 32.26 in Table 1 is a value we could not trace to a checked source; it is included only to show how sensitive g is to k.

Corollary 1.3 (spectral form)**.** *Let λ_0 be the bottom of the spectrum of the combinatorial Laplacian $4I - \mathcal{A}$ of the Cayley graph above, and ρ the spectral radius of simple random walk on it. Then*

$$\lambda_0 \;\geq\; 4 - \sqrt{16 - h_E^2} \;\geq\; \frac{h_E^2}{8} \;\geq\; 5.56 \times 10^{-39}, \qquad \rho \;\leq\; \sqrt{1 - h_E^2/16} \;\leq\; 1 - 1.39 \times 10^{-39},$$

and, under the hypothesis of Theorem 1.2 with $k = 29$, $\lambda_0 \geq 9.23 \times 10^{-18}$.

Proof. For an infinite graph of degree d with edge-isoperimetric constant i, Mohar [13] proved $\lambda_0 \geq d - \sqrt{d^2 - i^2}$ (see also Lyons–Peres [11], Ch. 6, for the form $\rho \leq \sqrt{1-(i/d)^2}$); and $d - \sqrt{d^2-i^2} \geq i^2/(2d)$. Take $d = 4$, and use $\rho = 1 - \lambda_0/4$ for the simple random walk on a regular graph. □

Known upper bounds. Guba [8] showed that the density is at least 3, so $h_E(F) \leq 1$. Belk and Brown [3] built finite sets with density tending to 3.5, so $h_E(F) \leq 1/2$. Guba [9] beat 3.5 (density > 3.5004), so $h_E(F) < 0.4996$. Hence

$$2.1 \times 10^{-19} \;\leq\; h_E(F) \;<\; 0.4996,$$

a gap of about 61 binary orders of magnitude (about 26 under the retraction hypothesis).

What is and is not new. The proof is the argument of [1] with every constant tracked. We change three things, all routine: (i) we use the Hilbert-space Lipschitz constant $1/\varepsilon$ for radial normalization in place of the Banach-space constant $2/\varepsilon$ used in [1] and its Lean code (Lemma 2.2); (ii) we evaluate the one non-explicit constant of the Lean construction, a curve-separation constant, by a certified computation (Proposition 4.1); (iii) we replace the finite set S of affine transports in [1] by $\{x_0, x_1\}$, using explicit normal-form words and averaging the transport errors instead of taking their maximum (Sections 3 and 5). No step of [1] fails and needs a new idea. The note's only contribution is that the constant is explicit: as far as we know it is the first explicit positive lower bound for an isoperimetric constant of F.

2 Two elementary inequalities

Lemma 2.1 (translation subadditivity)**.** *Let $A \subset F$ be finite and $w = s_1 \cdots s_\ell$ a word in $x_0^{\pm 1}, x_1^{\pm 1}$ with n_0 letters $x_0^{\pm 1}$ and n_1 letters $x_1^{\pm 1}$. Then $|wA \triangle A| \le n_0 |x_0 A \triangle A| + n_1 |x_1 A \triangle A|$.*

Proof. $|ghA \triangle A| \le |ghA \triangle gA| + |gA \triangle A| = |hA \triangle A| + |gA \triangle A|$, and $|x^{-1}A \triangle A| = |A \triangle xA|$. Induct on ℓ. □

For $\psi : F \to [-1, 1]$ and $h \in F$, cancellation over $hA \cap A$ gives [1, §2.3]

$$\left| \mathbb{E}_A \psi(h\,\cdot) - \mathbb{E}_A \psi \right| \le \frac{|hA \triangle A|}{|A|}, \qquad \mathbb{E}_A \psi = \frac{1}{|A|} \sum_{g \in A} \psi(g). \tag{1}$$

Lemma 2.2 (normalization in Hilbert space)**.** *In a real inner product space, for $x, y \neq 0$,*

$$\left\| \frac{x}{\|x\|} - \frac{y}{\|y\|} \right\| \le \frac{\|x - y\|}{\sqrt{\|x\|\|y\|}}.$$

Hence $x \mapsto -x/\|x\|$ is $1/\varepsilon$-Lipschitz on $\{\|x\| \ge \varepsilon\}$.

Proof. $\|x - y\|^2 - \|x\|\|y\| \left\| \frac{x}{\|x\|} - \frac{y}{\|y\|} \right\|^2 = \|x\|^2 + \|y\|^2 - 2\|x\|\|y\| = (\|x\| - \|y\|)^2 \ge 0$. (Equivalently, $x/\|x\| = P_B(x/\varepsilon)$ for $\|x\| \ge \varepsilon$, with P_B the nonexpansive nearest-point projection onto B; see [7].) □

Remark 2.3 (where this enters). The appendix of [1] defines $f(x) = -G(x)/\|G(x)\|$ with $\|G\| \ge \rho/4$ on B, and says "normalization on vectors of norm at least $\rho/4$ is Lipschitz with constant at most $8/\rho$". In the Lean code this is `normalizeNeg_lipschitz` in `AnalyticNormalization.lean` (the bound $2/\varepsilon$, proved for an arbitrary real normed space), used through `normalizedMap_lipschitz` in `AnalyticConstruction.lean`, where `displacementLipschitzConstant` $= 2\,\mathrm{Lip}(G)/(\rho/4)$. The space there is $H = L^2([0,1]; \mathbb{R}^2)$, an inner product space, so Lemma 2.2 applies with $\varepsilon = \rho/4$ and gives $\mathrm{Lip}(f) \le 4\,\mathrm{Lip}(G)/\rho$, half the formalized constant. The $2/\varepsilon$ bound is sharp in general normed spaces, which is presumably why the generic Lean lemma uses it. We have not changed the Lean code; Lemma 2.2 is a two-line identity.

3 The averaged variance inequality

Let H be a real Hilbert space with closed unit ball B, and let $f : B \to B$ satisfy $\|f(x) - f(y)\| \le L\|x - y\|$ and $\|f(x) - x\| \ge \delta > 0$. Put $\kappa = (\delta/L)^2$.

Dyadic layout. Identify a binary word w with the basic interval $[0.w, 0.w + 2^{-|w|}]$; then $I \cdot J$ is word concatenation. Fix $m \geq 1$, put $D = 2^m$, $r = m + 2$, and take the *parents*

$$P_u = u01 \qquad (u \in \{0,1\}^m),$$

ordered lexicographically. These are internal basic intervals of length 2^{-r} with pairwise positive gaps (between $u01$ and the next parent lies at least the interval $u1$). The descendants are $P_u \cdot P_v = u01v01$. The *reference pair* is $P_0 = 01 = [\frac{1}{4}, \frac{1}{2}]$, $Q_0 = 101 = [\frac{5}{8}, \frac{3}{4}]$. The proof of [1] works for any D internal, pairwise separated basic parents and any separated internal reference pair, since only the transport lemma and the covariance identity of [1, §2.1] are used. This layout and the averaged form below are not part of the Lean development, which uses the cells of depth $2D$ (`IntervalFamily.lean`) and the $\max_{h \in S}$ form; only the analytic map of Section 4 is taken from the formalization. Let $p(T)$ be the recursive coloring of [1] for the parents (P_u). Let $X_I(g)$ $(g \in F)$ be the color of the normalized restriction of $gT^{(n)}$ to I, for I in $\mathcal{I} = \{P_u\} \cup \{P_u P_v\} \cup \{P_0, Q_0\}$, at a level n that is admissible for every group element used below.

Transports. For each separated pair $I < J$ in $\mathcal{I} \setminus \{P_0, Q_0\}$, let $\varphi_{I,J} \in F$ carry I affinely onto P_0 and J onto Q_0 (transport lemma of [1, §2.1]; an explicit choice is in Section 5). By covariance [1, §2.1], $\langle X_I(g), X_J(g) \rangle = \langle X_{P_0}(\varphi_{I,J} g), X_{Q_0}(\varphi_{I,J} g) \rangle$, so with $\alpha = \mathbb{E}_A \langle X_{P_0}, X_{Q_0} \rangle$ and (1),

$$\big| \mathbb{E}_A \langle X_I, X_J \rangle - \alpha \big| \leq \epsilon(I,J) := \frac{|\varphi_{I,J} A \triangle A|}{|A|}. \tag{2}$$

The weighted pair family. Let $\mathcal{P}_m$ be the multiset of pairs, indexed by a row $u \in \{0,1\}^m$, consisting of *siblings* (uv, uv') for $v \neq v'$ (we write uv for $P_u P_v$; weight 1), *parents* $(P_w, P_{w'})$ for $w \neq w'$ (weight 1), and *mixed* pairs (uv, P_x) for v arbitrary and $x \neq u$ (weight 2). Ordered pairs are used throughout; each is transported in increasing order. The total weight is $W_m = 4D^2(D-1)$. Write $\bar{\epsilon}$ for the $\mathcal{P}_m$-weighted average of $\epsilon(I,J)$.

Proposition 3.1. *For every nonempty finite $A \subset F$,*

$$\kappa \;\leq\; \frac{4}{D} + 4\Big(1 - \frac{1}{D}\Big)\bar{\epsilon}.$$

Proof. For $g \in A$ put $m(g) = \frac{1}{D} \sum_u X_{P_u}(g)$ and $z_u(g) = \frac{1}{D} \sum_v X_{uv}(g)$. The recursion gives $X_{P_u} = f(z_u)$, so, as in the proof of the boundary proposition of [1, §2.3], $\delta^2 \leq \frac{L^2}{D} \sum_u \|z_u - m\|^2$ pointwise on A. Expand $\|z_u - m\|^2 = \|z_u\|^2 + \|m\|^2 - 2\langle z_u, m \rangle$ into D^{-2}-weighted inner products of colors. Diagonal terms are at most 1; the D nested mixed terms $\langle X_{uv}, X_{P_u} \rangle$ are at least -1. Every other term is a separated pair, and by (2) its average lies within $\epsilon(I,J)$ of α. The α-coefficients are $D(D-1) + D(D-1) - 2D(D-1) = 0$, so

$$\mathbb{E}_A \|z_u - m\|^2 \leq \frac{4}{D} + \frac{1}{D^2}\Big[\sum_{v \neq v'} \epsilon(uv, uv') + \sum_{w \neq w'} \epsilon(P_w, P_{w'}) + 2 \sum_v \sum_{x \neq u} \epsilon(uv, P_x)\Big].$$

Average over u, multiply by L^2, and note that the bracket summed over u has total weight $W_m = 4D^2(D-1)$: this gives $\delta^2 / L^2 \leq 4/D + (W_m / D^3)\,\bar{\epsilon}$, and $W_m / D^3 = 4(1 - 1/D)$. □

With $\max_{h \in S}$ in place of the average this is the boundary proposition of [1, §2.3]. Since a maximum dominates an average, the averaged form is legitimate. It matters here because the transport words have very different lengths.

Corollary 3.2. *Let $\bar{n}_i$ be the $\mathcal{P}_m$-weighted average of the number of letters $x_i^{\pm 1}$ in a word for $\varphi_{I,J}$, and let $\bar{M} \geq \max(\bar{n}_0, \bar{n}_1)$. If $\kappa > 4/D$, then for every nonempty finite A,*

$$\eta_0(A) + \eta_1(A) \;\geq\; \frac{\kappa - 4/D}{4(1-1/D)\,\bar{M}}.$$

Proof. By Lemma 2.1, $\epsilon(I,J) \leq n_0(I,J)\eta_0 + n_1(I,J)\eta_1$, so $\bar{\epsilon} \leq \bar{n}_0\eta_0 + \bar{n}_1\eta_1 \leq \bar{M}(\eta_0+\eta_1)$. Apply Proposition 3.1. □

Remark 3.3 (pairing). Each sibling term and each parent term in the expansion can be compared *directly* with a mixed term, through one transport between the two pairs, instead of comparing both with α. Then the error term is $2(1-1/D)\eta$ instead of $4(1-1/D)\eta$, while the diagonal and nested remainder stays $4/D$. In the S-form of [1] this halves the constant (sketch; not used below): $\max_{h\in S'} |hA \triangle A|/|A| \geq (\kappa - 4/D)/(2(1-1/D))$ for a suitable finite S'. In the $\{x_0, x_1\}$-form the direct transport costs, via the reference pair, at most the sum of the two reference transports. So with the word bounds used here the pairing gives nothing, and we do not use it.

4 The displacement map: explicit constants

4.1 The construction of the preprint

The appendix of [1] works in $H = L^2([0,1];\mathbb{R}^2)$. It uses the curve $\gamma(t) = a(t)v(t)$ with $v(t)(w) = (\cos\theta(t)w, \sin\theta(t)w)$ and

$$a(t) = \begin{cases} t/8, & t \leq 2 \\ \frac{5}{16} - \frac{(3-t)^2}{16}, & 2 < t < 3 \\ \frac{5}{16}, & t \geq 3 \end{cases} \qquad \theta(t) = \begin{cases} 0, & t \leq 1 \\ \frac{(t-1)^2}{2}, & 1 < t < 2 \\ t - \frac{3}{2}, & t \geq 2. \end{cases}$$

In the Lean structure `RegularCurveData`, the tube radius is $\rho = \min(1/1024, \sigma/8)$, where σ is any positive number with $\|\gamma(t) - \gamma(s)\| \geq \sigma$ whenever $|t-s| \geq 1/32$. The development obtains σ through `Classical.choose` from a compactness proof, so the formalized constant is not explicit. Everything else is explicit. (Lean files, all in `OAI/GroupTheory/Thompson/`: `TubeRegularCurve`, `AnalyticCorrection`, `CorrectionLocal`, `AnalyticConstruction`, `AnalyticNormalization`.) The correction G satisfies $\|G\| \geq \rho/4$, $G(x) = x$ for $\|x\| \geq 1/2$, and is Lipschitz with

$$\mathrm{Lip}(G) \leq \max\Big(1,\; 30 + \frac{7}{4\rho} + 3808\rho,\; 5 + \frac{7}{8\rho}\Big) \qquad (\texttt{CorrectionData.lipschitzConstant}).$$

Then $f = -G/\|G\|$ has $\|f(x) - x\| \geq \delta = 1/2$.

4.2 The separation constant

Proposition 4.1. $\sigma(1/32) := \inf_{|t-s|\geq 1/32} \|\gamma(t) - \gamma(s)\| = 1/256$.

Proof. Since $\langle v(s), v(t)\rangle = \mathrm{sinc}(\theta(t) - \theta(s))$,

$$\|\gamma(t) - \gamma(s)\|^2 = a_t^2 + a_s^2 - 2a_t a_s \,\mathrm{sinc}(\theta_t - \theta_s) = (a_t - a_s)^2 + 2a_t a_s\big(1 - \mathrm{sinc}(\theta_t - \theta_s)\big).$$

For $s < t \leq 1$ the curve is the segment $(t/8)v(0)$, so the distance is $(t-s)/8$. This gives $\sigma \leq 1/256$, with equality at $t - s = 1/32$. For the lower bound take $s < t$, $t - s \geq 1/32$, and split into cases:

(A1) $t \le 1$: the distance is $(t-s)/8 \ge 1/256$.

(A2) $s \le 0 < 1 < t$: here $a_s \le 0 < \frac{1}{8} \le a_t$. With $\min \operatorname{sinc} \ge -0.2173$ we get $\|\cdot\|^2 \ge (1-0.2173)a_t^2 \ge 0.7827/64$.

(A3) $0 \le s < t \le 2$: here a is linear with slope $1/8$, $a_s, a_t \ge 0$, and $1 - \operatorname{sinc} \ge 0$; so $\|\cdot\| \ge (t-s)/8$.

(A4) $s \ge 3$: here $\|\cdot\|^2 = 2(\frac{5}{16})^2(1 - \operatorname{sinc}(t-s)) \ge 2(\frac{5}{16})^2(1 - \operatorname{sinc}\frac{1}{32})$, since sinc decreases on $[0,\pi]$ and $|\operatorname{sinc}| \le 1/\pi < \operatorname{sinc}\frac{1}{32}$ beyond π.

(A5) $0 \le s \le 3 < 5 < t$: here $\theta_t - \theta_s \ge t - 3 \ge 2$, so $|\operatorname{sinc}| \le \frac{1}{2}$ and $\|\cdot\|^2 \ge a_t^2 + a_s^2 - a_t a_s \ge \frac{3}{4}a_t^2 = \frac{75}{1024} > \frac{3}{64}$.

(A6) $0 \le s \le 3$, $2 < t \le 5$: this region is compact, and we treat it by computer.

Cases (A2), (A4), (A5) have margins 801, 2.08 and 3072 over $(1/256)^2$; the inequality $\operatorname{sinc} \ge -0.2173$ is checked on $[1, 1/0.2173]$ by interval bisection (161 boxes), and outside that range it is trivial. In (A6) we use coordinates $(s, h = t - s)$ with $h \ge 1/32$ and run a branch-and-bound in Arb ball arithmetic (128-bit, outward rounding). On each box, a and θ are enclosed piece by piece. When $t \le 2$ we use the exact identity $a_t - a_s = h/8$. For $|q| \le 1$ we use the alternating-series enclosure $q^2/6 - q^4/120 \le 1 - \operatorname{sinc} q \le q^2/6$, and otherwise the Arb enclosure of $1 - \sin q/q$. A box is accepted when the enclosure of $\|\cdot\|^2$ lies strictly above $(1/256)^2$; 6462 boxes are accepted, and the smallest certified lower bound is $1.0039 \cdot (1/256)^2$. Box endpoints are rationals, entered as Arb balls containing them. □

The certificate is `cert.py`, part [A]. An independent re-implementation in `mpmath.iv` with a different region split (script `v1_sigma.py`) agrees.

4.3 The constant chain

With $\sigma = 1/256$, $\rho = \min(1/1024, 1/2048) = 1/2048$ and

$$\operatorname{Lip}(G) \le 30 + \tfrac{7}{4} \cdot 2048 + \tfrac{3808}{2048} = \frac{231415}{64} = 3615.859375, \qquad \varepsilon = \frac{\rho}{4} = \frac{1}{8192}.$$

By Lemma 2.2, $L = \operatorname{Lip}(G)/\varepsilon = 29621120$, against 59242240 with $2/\varepsilon$. Hence

$$\kappa = \Big(\frac{1/2}{29621120}\Big)^2 = 2.8493 \times 10^{-16} \qquad (\text{with } 2/\varepsilon :\ 7.1232 \times 10^{-17}).$$

Substituting $\sigma = 1/256$ for the `Classical.choose` value is legitimate because every Lean lemma downstream of `RegularCurveData` is proved for an arbitrary separation constant satisfying the separation property, and Proposition 4.1 supplies that property for $1/256$. That proposition is computer-assisted and not formalized.

4.4 Maps from retractions

Lemma 4.2. *If $R : B \to S$ is a k-Lipschitz retraction, then $f = -R$ is k-Lipschitz on B and $\|f(x) - x\| \ge 2/(k+2)$.*

Proof. Let $t = \|x\|$. For $x = 0$, $\|R(0)\| = 1$. Otherwise put $y = x/t \in S$, so $R(y) = y$. Then $\|R(x) + x\| \ge \|y + x\| - \|R(x) - R(y)\| \ge (1+t) - k(1-t)$, and also $\|R(x) + x\| \ge 1 - t$. The maximum of these two lower bounds is at least its value $2/(k+2)$ at $t = k/(k+2)$, where they cross. □

Thus $\kappa_k = (2/(k(k+2)))^2$, which gives Theorem 1.2 once Section 5 supplies $\bar{M}$.

5 Transport words in x_0, x_1

Tree pairs and normal form. A finite rooted binary tree T is identified with its set of leaves, a set of binary words whose intervals tile $[0,1]$. Let N be its number of carets, so it has $N+1$ leaves. For two trees with the same N, let $(T_- \to T_+) \in F$ be the element mapping the k-th leaf interval of T_- affinely onto the k-th leaf interval of T_+. Let R_N be the right vine. Following Cannon–Floyd–Parry [6, §2], the *exponent* e_k of the k-th leaf ($0 \le k \le N$) is the length of the maximal chain of left edges ending at that leaf whose edges do not hang from the right spine; that is, an edge from a vertex 1^j is not counted. In our convention $hg = h \circ g$, with $x_{j+1} = x_0^{-1} x_j x_0$,

$$(R_N \to T) = x_0^{e_0} x_1^{e_1} \cdots x_N^{e_N} = x_0^{e_0}\, x_1^{e_1}\, x_0^{-1} x_1^{e_2}\, x_0^{-1} \cdots x_0^{-1} x_1^{e_n}\, x_0^{n-1}, \tag{3}$$

where $n = \max\{k \ge 1 : e_k > 0\}$ (the telescoped word is $x_0^{e_0}$ if no such k exists). The certificate checks (3) in exact arithmetic on 400 random trees with up to 9 carets, by verifying that the word maps every leaf of R_N affinely onto the corresponding leaf of T. In the proof below we only need that the final words transport the right intervals, and the certificate checks this directly for every pair at $m \le 3$, a sample at $m = 4$, and random pairs with $m \le 64$. We call $P(T)$ the right-hand word in (3). Then $(T_- \to T_+) = P(T_+)P(T_-)^{-1}$, and we freely reduce the concatenation.

Two facts follow from the definition. First, every caret has exactly one left edge, and each left edge lies in exactly one maximal left chain; the uncounted edges are those hanging from the s spine vertices $1^0, \dots, 1^{s-1}$, where s is the depth of the last leaf 1^s. Hence

$$\textstyle\sum_{k=0}^{N} e_k = N - s. \tag{4}$$

Second, $e_N = 0$. The penultimate leaf is $1^{s-1}01^j$: if $j \ge 1$ it ends in 1, and if $j = 0$ its only left edge hangs from 1^{s-1}. Either way $e_{N-1} = 0$, so $n \le N-2$.

The transport. For a pair $I < J$ in $\mathcal{P}_m$, let T_{dom} be the minimal tree having I and J as leaves. Its internal vertices are exactly the proper prefixes of I and J. Its leaves split into a left gap (before I), a middle gap and a right gap. The reference tree $\{00, 01, 100, 101, 11\}$ has gaps $\{00\}, \{100\}, \{11\}$. Let T_{ran} be obtained from it by repeatedly splitting the last leaf of each gap until each gap has as many leaves as the corresponding gap of T_{dom}. Then $\varphi_{I,J} = (T_{\mathrm{dom}} \to T_{\mathrm{ran}})$ carries I onto 01 and J onto 101 affinely.

Lemma 5.1 ((F1))**.** *All three gaps of T_{dom} are nonempty, so no padding of T_{dom} is needed, and*

$$N(\varphi_{I,J}) = |I| + |J| - \mathrm{lcp}(I,J) - 1.$$

Proof. The left gap is empty only if I is the leftmost leaf $0^{|I|}$; the right gap is empty only if $J = 1^{|J|}$; the middle gap is empty only if I and J are adjacent. Our intervals are internal and separated, so none of these happens. The proper prefixes of I and J number $|I| + |J|$ counted with multiplicity; the common ones are those of length $0, \dots, \mathrm{lcp}(I,J)$, and $\mathrm{lcp} < \min(|I|, |J|)$ because the intervals are not nested. So there are $|I| + |J| - \mathrm{lcp} - 1$ carets. □

Lemma 5.2 ((F2))**.** *Let $e_0^{\mathrm{d}}, e_0^{\mathrm{r}}$ be the leftmost exponents and $s_{\mathrm{d}}, s_{\mathrm{r}}$ the depths of the last leaves of $T_{\mathrm{dom}}, T_{\mathrm{ran}}$, and $N = N(\varphi_{I,J})$. The freely reduced word $w = P(T_{\mathrm{ran}})P(T_{\mathrm{dom}})^{-1}$ satisfies*

$$n_1(w) \le (N - s_{\mathrm{r}} - e_0^{\mathrm{r}}) + (N - s_{\mathrm{d}} - e_0^{\mathrm{d}}), \qquad n_0(w) \le e_0^{\mathrm{r}} + e_0^{\mathrm{d}} + 2N - 6,$$

with $e_0^{\mathrm{d}} = \mathrm{lz}(I)$, the number of leading zeros of I, and $1 \le e_0^{\mathrm{r}} \le 2$.

Proof. By (3), $P(T)$ has $\sum_{k\ge1} e_k = N - s - e_0$ letters $x_1^{\pm1}$, by (4). It has $e_0 + 2(n-1)$ letters $x_0^{\pm1}$ (or e_0 if $n = 0$): e_0 leading ones, $n-1$ interior x_0^{-1}, and the tail x_0^{n-1}. Write $a = n_{\mathrm{r}} - 1$ and $b = n_{\mathrm{d}} - 1$. In $P(T_{\mathrm{ran}})P(T_{\mathrm{dom}})^{-1}$ the tail x_0^a meets the head x_0^{-b} and freely reduces to x_0^{a-b}. The x_0-count is therefore at most $e_0^{\mathrm{r}} + e_0^{\mathrm{d}} + a + b + |a-b| = e_0^{\mathrm{r}} + e_0^{\mathrm{d}} + 2\max(a,b) \le e_0^{\mathrm{r}} + e_0^{\mathrm{d}} + 2(N-3)$, using $n \le N-2$; when $n = 0$ the bound holds since $N \ge 3$. The leftmost leaf of T_{dom} is $0^{\mathrm{lz}(I)+1}$, and its left chain hangs from the root, so $e_0^{\mathrm{d}} = \mathrm{lz}(I)$. The leftmost leaf of T_{ran} is 00 or 000. □

Lemma 5.3 ((F2′), sharper)**.** *Let $g = g(J)$ be the number of zeros in the word J. If J ends in* 1 *(true for every interval of the layout), then*

$$n_0(w) \le 2N - 2g + e_0^{\mathrm{d}} - 2, \qquad n_1(w) \le 2N - g - 3 - e_0^{\mathrm{d}}.$$

Proof. The leaves of T_{dom} to the right of J are the words $J_{<i}1$ for the positions i with $J_i = 0$; none of them is a prefix of I, because I lies to the left of J. So the right gap has g leaves, J has index $N-g$, and all of these leaves, and J itself, end in 1 and have exponent 0. In T_{ran} the leaves from index $N-g$ on are $101, 110, 1110, \dots, 1^g0, 1^{g+1}$ (for $g = 1$: 101, 11), all of exponent 0, so $s_{\mathrm{r}} = g+1$. Hence $n_{\mathrm{d}}, n_{\mathrm{r}} \le N - g - 1$, and the proof of Lemma 5.2 gives $n_0 \le e_0^{\mathrm{r}} + e_0^{\mathrm{d}} + 2(N-g-2)$ and $n_1 \le (N - g - 1 - e_0^{\mathrm{r}}) + (N - s_{\mathrm{d}} - e_0^{\mathrm{d}})$. Finally use $1 \le e_0^{\mathrm{r}} \le 2$ and $s_{\mathrm{d}} \ge 1$. □

Lemma 5.2 is Lemma 5.3 without the gain from the right gap (that is, with $g = 1$). Both are checked pointwise for every pair at $m \le 4$ and for 8500 random pairs at $m \in \{8, 16, 24, 40, 64\}$ (certificate, part [C]). A separate implementation (`v2_words.py`) also checks Lemmas 5.1 and 5.2, and another (`r1_F2sharp.py`) checks Lemmas 5.1 and 5.3.

Lemma 5.4 (closed-form averages)**.** *Let $\lambda = \frac{D-1-m}{D-1}$, let $a_k = \#\{u : \mathrm{lz}(u01) \ge k\}$ (so $a_k = D2^{-k}$ for $k \le m$, $a_{m+1} = 1$, and $a_k = 0$ beyond), and let $E_P = \sum_{k=1}^{m+1}\big(1 - \frac{(D-a_k)(D-a_k-1)}{D(D-1)}\big)$. The $\mathcal{P}_m$-averages of N, $\mathrm{lz}(I)$ and $g(J)$ are*

$$\bar N = \frac{11r}{4} - 1 - \lambda, \qquad \bar e_0 = \frac{1 + 3E_P}{4}, \qquad \bar g = \frac{3m+4}{4}.$$

Consequently, by Lemma 5.2, $\bar M(m) = \max(2\bar N + \bar e_0 - 4,\ 2\bar N - 2)$, and by Lemma 5.3, $\bar M'(m) = \max(2\bar N - 2\bar g + \bar e_0 - 2,\ 2\bar N - \bar g - 3 - \bar e_0)$, are admissible in Corollary 3.2.

Proof. By Lemma 5.1: siblings $u01v01 < u01v'01$ give $N = 3r - 1 - \mathrm{lcp}(v, v')$; parents give $N = 2r - 1 - \mathrm{lcp}(w, w')$; mixed pairs give $N = 3r - 1 - \mathrm{lcp}(u, x)$. For distinct uniform $v, v' \in \{0,1\}^m$, $\mathbb{E}\,\mathrm{lcp} = \sum_{k=1}^m \Pr[\mathrm{lcp} \ge k] = \sum_k \frac{D2^{-k}-1}{D-1} = \lambda$; weight the three cases $1:1:2$. For $\bar e_0$: the lower interval of a sibling pair is $u01\cdots$, with $\mathbb{E}\,\mathrm{lz} = \sum_k a_k/D = 1$. The lower of $P_w, P_{w'}$ is the lexicographically smaller one, and its leading-zero count is the larger of the two, which has expectation E_P. A mixed pair is ordered by comparing u with x, with the same expectation. For $\bar g$: for distinct uniform w, w', the larger has a 1 where the smaller has a 0 at the first difference and is otherwise identically distributed, so $\mathbb{E}z(\max) = (m-1)/2$, where z counts zeros. The upper intervals are $u01\max(v,v')01$ (siblings, $\mathbb{E}g = m + \frac{3}{2}$), $\max(w,w')01$ (parents, $\frac{m+1}{2}$), and, with probability $\frac12$ each, $x01$ or $u01v01$ with $u = \max$ (mixed, $\frac{3m+4}{4}$). The weighted mean is $\frac{3m+4}{4}$. □

The certificate compares all three closed forms with exhaustive enumeration for $m \le 4$ (equality of rationals). For instance $\bar M(59) = 329.5 + 2(1-\lambda)$, i.e. 329.5 up to 2×10^{-16}, and $\bar M'(59) = 281.75$. The exact averages $\bar n_0, \bar n_1$ at $m \le 6$ lie well below both bounds: at $m = 6$ they are 28.15 and 27.72, against $\bar M = 38.19$ and $\bar M' = 30.18$. Extrapolating the growth rates (about 4.55 per unit of m for $\bar n_1$, against 4.75 for $\bar M'$) suggests that exact counts would gain only about a further 5% at $m = 59$; we do not use this.

k	κ_k	optimal m	$\bar{M}(m)$	$g(k)$ [$g'(k)$ with $\bar{M}'$]
3	1.78×10^{-2}	11	65.51	6.042×10^{-5} [7.362×10^{-5}]
4.58	4.40×10^{-3}	13	76.50	1.280×10^{-5} [1.548×10^{-5}]
6	1.74×10^{-3}	15	87.50	4.612×10^{-6} [5.547×10^{-6}]
10	2.78×10^{-4}	18	104.0	6.311×10^{-7} [7.544×10^{-7}]
20	2.07×10^{-5}	22	126.0	3.910×10^{-8} [4.648×10^{-8}]
28.99	4.956×10^{-6}	24	137.0	8.6085×10^{-9} [1.0211×10^{-8}]
29	4.949×10^{-6}	24	137.0	8.5964×10^{-9} [1.0197×10^{-8}]
32.26	3.275×10^{-6}	24	137.0	5.540×10^{-9} [6.572×10^{-9}]
33	2.998×10^{-6}	25	142.5	5.051×10^{-9} [5.986×10^{-9}]
50	5.92×10^{-7}	27	153.5	9.152×10^{-10} [1.083×10^{-9}]
100	3.84×10^{-8}	31	175.5	5.211×10^{-11} [6.149×10^{-11}]

Table 1: The conditional bound $g(k)$ of Theorem 1.2; in brackets, $g'(k)$ from Theorem 6.1. Lower bounds for $k_0(H)$: 4.5 [7], 4.58 [10] (quoted in [12]). Upper bound: 28.99, quoted from [10, 2] (neither read by us).

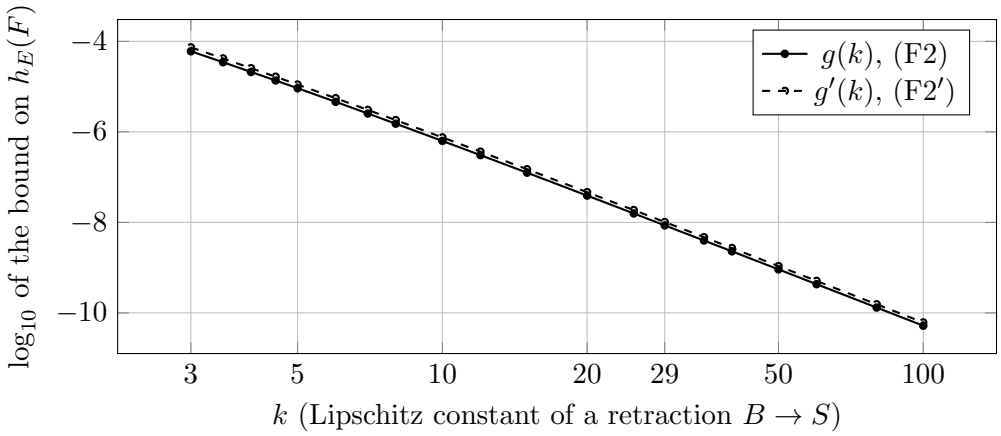


Figure 1: Conditional lower bounds as functions of the retraction constant. For comparison, the unconditional bound is $\log_{10} c_\star = -18.68$.

6 Proofs of the theorems

Proof of Theorem 1.1. Take f from Section 4 with $\delta = 1/2$ and $L = 29621120$, so $\kappa = 2.8493 \times 10^{-16}$. In Corollary 3.2 take $\bar{M} = \bar{M}(m)$ and maximize over m; the maximum is at $m = 59$. The resulting exact rational is $c_\star = 2.10918\ldots \times 10^{-19}$ (Arb). □

Theorem 6.1. *With $\bar{M}'(m)$ in place of $\bar{M}(m)$, $h_E(F) \geq 2.4666 \times 10^{-19}$ (at $m = 59$), and under the hypothesis of Theorem 1.2, $h_E(F) \geq g'(k)$ with $g'(29) = 1.0197 \times 10^{-8}$.*

Lemma 5.3 is proved here, checked by the certificate, and independently re-checked (exhaustively for $m \leq 6$, on random pairs up to $m = 100$, with whole-map transport checks). The bounds of Lemma 5.3 are attained for some pairs.

Proof of Theorem 1.2. Lemma 4.2 and Corollary 3.2 with $\bar{M}(m)$. □

factor	$\log_2$	origin
$\delta^2 = 1/4$	-2.00	displacement of $f = -G/\|G\|$
$\varepsilon^2 = (\rho/4)^2$, $\rho = 1/2048$	-26.00	$\rho = \sigma/8$, $\sigma = \frac{1}{256} = \frac{1}{8} \cdot \frac{1}{32}$ (speed $\times$ scale)
[with $2/\varepsilon$ as in Lean]	$[-28.00]$	Banach-space normalization constant
$\mathrm{Lip}(G)^{-2}$, $\mathrm{Lip}(G) = 3615.86$	-23.64	dominated by the cutoff term $7/(4\rho) = 3584$
$\kappa = (\delta/L)^2$	-51.64	
$(1 - 4/(D\kappa))/(4(1 - 1/D))$	-2.04	variance constant 4; $D = 2^{59}$
$1/\bar{M}$, $\bar{M} = 329.5$	-8.36	word length of transports, $\approx 5.5m$
$h_E \geq c_\star$	-62.04	(-61.81 with (F2′), -64.09 with $2/\varepsilon$)

Table 2: Ledger for Theorem 1.1.

7 Where the constant was lost, and the limits of the method

Table 2 itemizes the loss. About 52 of the 62 binary orders come from the analytic map: the squared ratio $(\delta/L)^2$ with $L \approx 7/\rho^2 \approx 2^{24.8}$. This in turn comes from the tube radius $\rho = 1/2048$, set by the slow speed $1/8$ of the curve, the separation scale $1/32$, and the safety factor 8 in $\rho = \sigma/8$. The group-theoretic part, from the transports, costs about 10 binary orders, of which 8.4 are the average word length; it is the only part that depends on F.

Ceilings. (i) For every L-Lipschitz $f : B \to B$ the minimal displacement is at most $1 - 1/L$ (Goebel; see [5]), so $\delta/L \leq 1/4$ and $\kappa \leq 1/16$. Inserted into Corollary 3.2 with the present layout, this caps the method at $h_E \geq G(1/16) = 2.51 \times 10^{-4}$ ($m = 9$). In the S-form of [1] the cap is $\kappa/4 \leq 1/64$ for $\max_{h \in S} |hA \triangle A|/|A|$. (ii) Under the retraction hypothesis, $\kappa_k \leq (2/15)^2$ at $k = 3$, the universal lower bound for k_0, so $g(k) \leq g(3) = 6.04 \times 10^{-5}$. (iii) As long as each pair is compared with a reference pair through a single transport, the average word length grows linearly in m, while $m \geq \log_2(4/\kappa)$ is forced. So the $\{x_0, x_1\}$-form always loses a factor of order $\log(1/\kappa)$ relative to the S-form. Compared with the known upper bound 0.4996, even ceiling (i) is about 11 binary orders short.

How one could do better. The real leverage is in κ. A different curve (faster speed, larger separation scale), or a Lipschitz map with better displacement-to-Lipschitz ratio in Hilbert space, would help most; under the retraction hypothesis this is the passage from 2.1×10^{-19} to 8.6×10^{-9}. On the group side, exact word counts (rather than the bounds of Lemmas 5.2–5.3) and better reference pairs could each gain a constant factor. We have not pursued these.

Reproducibility

`cert.py` (Python 3, `python-flint` 0.9) recomputes, in about 15 seconds on one core: [A] Proposition 4.1; [B] the constant chain, in exact rationals; [C] the word-length facts and closed forms (exhaustive for $m \leq 4$, random up to $m = 64$, including the transport property of the words); [D] all bounds in Theorems 1.1, 1.2, 6.1 and Table 1, as exact rationals printed through Arb; [E] the ceilings, the ledger and Corollary 1.3. Its output is saved as `cert_output.txt`. Separately written scripts check the same ingredients: `s1_sigma_paper_curve.py` and `v1_sigma.py` (the latter in `mpmath.iv`) check Proposition 4.1; `s4_wordlength_R.py`, `v2_words.py` and `v3_avg.py` check (F1), (F2) and the closed forms; and `v4_constants.py` reproduces the (F2)-based values 2.1092×10^{-19},

5.1026×10^{-20} and 8.5964×10^{-9}; `r1_F2sharp.py` checks (F1), (F2′), the closed-form averages and the transport property with its own data structures.

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